LINING YAN

lining.yan@yale.edu | +1 574-310-7057 | New Haven, CT, USA

Research Interests: Human-Centered AI · Interactive & Adaptive Agents · Multimodal Learning · Embodied AI

EDUCATION

Yale University — M.S. in Computer Science *New Haven, CT | Sep 2026 – Expected 2027*

Research Focus: Artificial Intelligence, Robot Learning, Reinforcement Learning, and Embodied & Interactive AI
Selected Coursework: Intelligent Robotics, Advanced Robotic Mechanisms, Sequential Decision Making, Advances in Frontier AI Models, Introduction to Artificial Intelligence

Kean University — B.S. in Computer Science, Minor in Applied Mathematics *Union, NJ, USA | 2022 – 2026*
GPA: 3.92/4.0 · Summa Cum Laude · Dean's List (2023, 2024, 2025)
Selected Coursework: Algorithms & Data Structures, Advanced Machine Learning, Generative AI, Big Data Analytics, Numerical Analysis.

Stanford University — Summer Session *Stanford, CA, USA | Jun – Aug 2024*
GPA: 4.12/4.3. Coursework: Introduction to High-Performance Computing, HPC-AI Summer Seminar Series, Data Mining and Analysis, Introduction to Stochastic Processes.

SKILLS

Programming: Python, Java, SQL, NumPy, pandas, scikit-learn
AI & Machine Learning: PyTorch, Transformers, Deep Learning, NLP, Multimodal Learning, Reinforcement Learning
Tools & Computing: Git, Linux, Docker, High-Performance Computing (HPC)
Languages: English (Fluent), Mandarin Chinese (Native)

RELEVANT EXPERIENCE

Yale University — Inquisitive Robotics Lab, Graduate Researcher (Advisor: Prof. Tesca Fitzgerald) *New Haven, CT | Sep 2026 – Present*

- Exploring human-centered interactive AI for adaptive agents, with a focus on how intelligent systems can recognize uncertainty or failure, seek human input when needed, and improve through corrective feedback.
- Examining uncertainty-aware multimodal and vision-language-action models as foundations for help-seeking, interactive learning, and post-failure adaptation in real-world tasks.

University of Notre Dame — Center for Research Computing, Research Intern (Advisor: Dr. Samuel Rund) *Notre Dame, IN | Jul – Aug 2026*

- Developed an agentic AI system for extracting structured scientific evidence from multimodal research documents, integrating reasoning over text, tables, and figures.
- Designed adaptive multi-step retrieval and verification workflows that determine when additional evidence is needed, recover from incomplete or inconsistent outputs, and provide grounded answers across large scientific corpora.

National Research Institute — Research Intern (Advisor: Dr. Guoxin Kang) *Beijing, China | Jan – Jun 2026*

- Researched efficient multimodal retrieval, developing non-autoregressive routing methods for large-scale cross-modal search.
- Formulated semantic routing as an entropy-regularized optimal transport problem and developed Sinkhorn-balanced routing with local candidate filtering and exact similarity reranking.

Wenzhou-Kean University — NLP Research Group, Undergraduate Researcher (Advisor: Dr. Baha Ihnaini) *Zhejiang, China | 2023 – 2025*

- Conducted two years of advised research on AI-generated text detection and interpretability, spanning global/local linguistic feature analysis, ensemble modeling, and semi-supervised training under limited labels.
- *Primary research output:* Multi-Level Text Feature Analysis for Detecting AI-Generated Content and Semi-Supervised Artificial Intelligence Text Detection with RoBERTa.

PUBLICATIONS & MANUSCRIPTS

Multi-Level Text Feature Analysis for Detecting AI-Generated Content *CSCI 2025*

- **First Author.** Analyzed AI-generated vs. human-written abstracts using global and local text features; developed a fine-tuned BERT + Random Forest ensemble achieving 99.9% (abstract-level) and 93.5% (sentence-level) F1. Used SHAP analysis to identify perplexity and log-probability as key discriminative features.

Bridging the Modality Gap that Hurts Cross-Modal Retrieval *Under Review, VLDB 2026*

- **First Author.** Proposed a non-autoregressive multimodal retrieval framework using Sinkhorn-balanced semantic buckets, a 2-layer MLP router, bucket-local CE filtering, and exact cosine reranking. Achieved R@1 = 53.6% (MS-COCO) and R@5 = 46.7% (Visual News) while exploring ~1.4% of the corpus; router latency of 0.15 ms/query, 13× faster end-to-end inference than GENIUS on Visual News.

GARDS-RAG: Directed Acyclic Graph-Edge State Routing *In preparation for NAACL 2027*

- **Primary Researcher.** Developed a graph-structured RAG framework that preserves the original question as a global anchor while routing dependency-local reasoning states along DAG edges for multi-step retrieval and generation.
- Designed dependency-conditioned evidence reranking and node-level generation using direct-predecessor states, reducing irrelevant reasoning history while preserving global task intent; evaluated across MuSiQue, 2WikiMultiHopQA, and HotpotQA with multiple LLM and retrieval backbones, outperforming iterative and graph-structured RAG baselines.

MCP-X: An Ultra-Compact CNN for Rice Disease Classification *AgriEngineering 2025, 7(11), 359*

- **Second Author.** Designed a 0.21M-parameter CNN for real-time rice disease classification in resource-constrained settings; achieved 98.93% validation accuracy on PlantVillage, outperforming MobileNetV2 with far fewer parameters.

Pattern-Centric Detection of AI-Generated Text: An Interpretable and Data-Efficient Expert System *In preparation*

- **Author (Senior Capstone).** Built a pattern-centric expert system — a KMeans-derived Generative Pattern Library over multi-granularity linguistic features paired with an Explainable Boosting Machine (EBM) — achieving AUROC above 0.99 using only 44–50 labeled samples across five benchmarks (DAIGT, HC3, LLMDetect, MAGE, MultiModel), retaining over 95% of full-model performance at ~1/100th the labeling cost and outperforming zero-shot DetectGPT on all five.
- Discriminative across 27+ LLM families and adversarial perturbations without white-box model access (applicable to closed-source LLMs); every prediction carries an auditable SHAP feature attribution, and a modular pattern library enables rapid domain adaptation without full pipeline retraining.

Semi-Supervised Artificial Intelligence Text Detection with RoBERTa *In preparation*

- **Lead Researcher.** Developed a semi-supervised RoBERTa framework achieving up to 97.79% F1 on the HC3 dataset using only 1% labeled data, outperforming BERT, ELECTRA, ModernBERT, and baseline RoBERTa.

AI-Enhanced Echocardiography: Comparative Analysis of Image Quality Using ESRGAN

- **Lead Researcher.** Applied generative AI to improve echocardiographic image quality and clinical-metric accuracy across Original, CycleGAN, ESRGAN, and a novel ESRGAN-Denoise pipeline; ESRGAN-Denoise achieved the best segmentation (Dice 0.9670, IoU 0.9362) and clinical alignment, indicating strong potential for clinical deployment.

TEACHING & CAMPUS SERVICE

Wenzhou-Kean University — Peer Tutor *Zhejiang, China | Oct 2023 – Dec 2025*

- Provided one-on-one guidance on learning methods and course material to 80+ students; met regularly with academic support center staff to track student progress.

Wenzhou-Kean University — Computer Laboratory Assistant *Zhejiang, China | Oct 2023 – Dec 2025*

- Checked and maintained laboratory equipment; conducted weekly equipment test records and reports.

HONORS & AWARDS

- Best Poster Award, Student Research Day (2024)
- Scholarship for Research and Innovation (2024)
- Third Prize, AI Shark Tank Contest — AI & University Life (2024)
- Dean's Second Prize Scholarship (2023, 2024, 2025)
- Zhejiang Provincial Government Scholarship (2024, 2025)
- Honorable Mention, US Collegiate Mathematical Contest in Modeling (2024)
- Outstanding Graduate of Zhejiang Province (2026)